

THESEUS IN THE GRAPH: LEARNING TO TRAVERSE KNOWLEDGE GRAPHS FROM NATURAL-LANGUAGE QUESTIONS

Eduin E. Hernandez, Luis F. Garcia, Nurassyl Askar, and Stefano Rini

National Yang-Ming Chiao-Tung University (NYCU), Taiwan

ABSTRACT

Knowledge graphs (KGs) provide a structured representation of facts and are a natural substrate for reasoning across large repositories of information. A central challenge is *multi-hop reasoning*, where answering a question requires composing multiple relations to traverse from a specified start entity to the correct answer. Despite its importance, most existing approaches remain focused on atomic link prediction or single-relation queries, leaving the task of navigating KGs to answer truly compositional, natural-language questions underexplored. We present a new task, *question-conditioned navigation* in KGs: given a KG, a natural-language question, and a start node, an agent must reach the answer via multi-hop paths. To this end, we (i) curate dedicated QA–KG resources designed to support multi-hop navigability; (ii) analyze their structural properties to characterize traversal difficulty; and (iii) present a BERT–KG framework that learns a shared latent space between natural-language questions and KG embeddings to condition path selection. We further adapt established baselines such as MINERVA, MultiHopKG, and SQUIRE to operate on full-question embeddings rather than single-relation inputs, thereby enabling compositional, multi-relation questions beyond atomic link prediction. Finally, we propose evaluation protocols that jointly assess answer accuracy and hop-length coverage.

Index Terms— Knowledge Graphs, Question Answering, Multi-hop Reasoning, Question-Conditioned Navigation

1. INTRODUCTION

Knowledge graphs (KGs) provide a structured representation of facts and offer a natural substrate for reasoning with large language models (LLMs) [1, 2, 3]. If equipped with the ability to navigate a KG, an LLM could answer complex questions by composing multi-hop relational chains across entities—an ability with wide-ranging applications. However, most of the existing literature has concentrated on *link prediction*, i.e., recovering missing entities in atomic queries [4, 5]. Even work explicitly labeled as “multi-hop reasoning” often reduces to

justifying a single relation by searching for supporting paths, rather than reasoning over full natural-language questions. As a result, the challenge of true *multi-hop navigation* in KGs remains largely overlooked. In this work, we take a first step toward filling this gap by studying the question–answering setting. More specifically, we introduce *question-conditioned navigation*, in which an agent must traverse a KG from a specified start node to the node corresponding to the answer of a natural-language question.

Related Work: Classical KG research has emphasized *knowledge graph completion*, typically via embedding-based scoring functions for link prediction [6, 7, 8, 9, 10]. Path-finding methods such as DeepPath [11] and MINERVA [12] frame reasoning as reinforcement learning, where an agent walks the graph from a head entity toward a tail entity. Extensions like MultiHopKG [13] introduce reward shaping using knowledge graph embeddings, while sequence-to-sequence models such as SQUIRE [14] plan entire paths directly in sequence space. Despite these innovations, most approaches still operate on atomic queries and evaluate only end-node prediction, falling short of question-driven, compositional reasoning.

Our contributions: We address this gap by introducing the task of *question-conditioned navigation*: given a KG, a natural-language question q , and a start node e_s , an agent must traverse a multi-hop path to reach the answer node e^* . An n -hop question requires composing a length- n relational chain from e_s to e^* , integrating $n-1$ intermediate entities. Unlike link prediction, the objective is not to infer missing links, but to *navigate and reason* over an existing graph. Our contributions are:

- **Datasets:** we release QA–KG resources curated for multi-hop navigability and alternative traversal routes. We also report structural statistics of the underlying KGs to quantify path diversity and traversal difficulty.
- **Multimodal alignment:** we present a BERT–KGE framework that learns a shared latent space between question embeddings and KG embeddings, conditioning path selection on language and KG neighborhood.
- **Adapted baselines & evaluation:** we modify MINERVA [12], MultiHopKG [13], and SQUIRE [14] to operate on full-question embeddings, and we propose evaluation protocols reporting both *answer accuracy* and *hop-length coverage*.

This work is partially funded by the NSTC grant number 111-2221-E-A49-068-MY3 and by MARC, the MediaTEK Advanced Research Center with grant number 113UA10035.

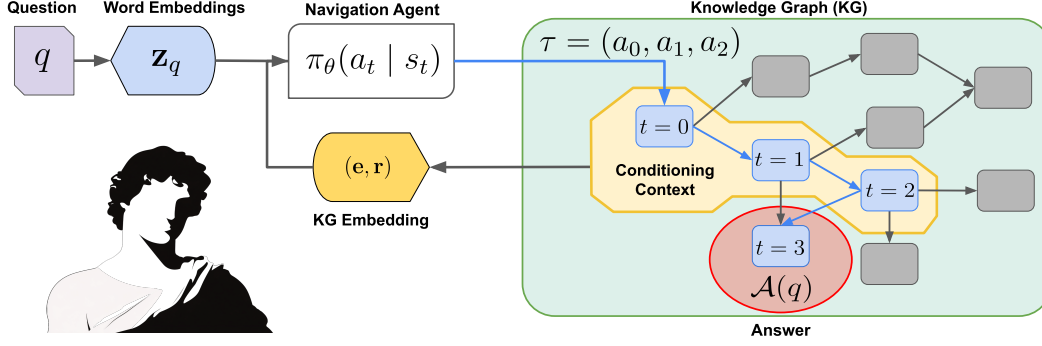


Fig. 1. Theseus in the graph— *Question-conditioned multi-hop navigation in a KG*. Given a question q with embedding $\mathbf{z}_q$, the policy $\pi_\theta(a_t | s_t)$ selects a set of actions a_0, a_1, a_2 to reach an answer $\mathcal{A}(q)$. At each step, it observes a graph embedding $(\mathbf{e}, \mathbf{r})$ of conditioning context. The blue path τ represents one successful trajectory, the other gray paths represent the other alternatives.

2. PROBLEM FORMULATION

2.1. Preliminaries

KG and Multi-Hop Reasoning: A KG is defined by a set of entities $\mathcal{E}$ and a set of relations $\mathcal{R}$. Each directed edge in the graph can be represented as a triple (h, r, t) , where $h, t \in \mathcal{E}$ are the head and tail entities, and $r \in \mathcal{R}$ is the relation connecting them. We denote by $\mathcal{T}$ the set of all such triples. Given a start entity $e_s \in \mathcal{E}$ and a natural-language question q , multi-hop reasoning requires composing a sequence of relational edges $r_1, r_2, \dots, r_n$ to reach an answer entity e^* . Let $e_0 = e_s$ and $e_n = e^*$; then a valid reasoning path P is defined as:

$$P = \{(e_0, r_1, e_1), (e_1, r_2, e_2), \dots, (e_{n-1}, r_n, e_n)\}.$$

The integer n specifies the number of hops in the reasoning process, with intermediate nodes $e_1, \dots, e_{n-1}$ along the path. Multi-hop navigation thus amounts to selecting and executing such a path in response to a given question.

KG Embedding: Let d_{KG} be the embedding dimension, and let the entity and relation embedding matrices be $\mathbf{E} \in \mathbb{R}^{|\mathcal{E}| \times d_{KG}}$ and $\mathbf{R} \in \mathbb{R}^{|\mathcal{R}| \times d_{KG}}$. For $h, t \in \mathcal{E}$ and $r \in \mathcal{R}$, denote embeddings by $\mathbf{e}_h := \mathbf{E}_{[h, :]}$, $\mathbf{e}_t := \mathbf{E}_{[t, :]}$, and $\mathbf{r} := \mathbf{R}_{[r, :]}$. Triples are scored by

$$s(h, r, t) = g(\mathbf{e}_h, \mathbf{r}, \mathbf{e}_t),$$

with some common choices being:

- *TransE* [6]:

$$s_{\text{TransE}}(h, r, t) = -\|\mathbf{e}_h + \mathbf{r} - \mathbf{e}_t\|_p, \quad p \in \{1, 2\}.$$

- *RotatE* [7]:

$$s_{\text{RotatE}}(h, r, t) = -\|\mathbf{h} \circ \mathbf{r} - \mathbf{t}\|_1, \quad |\mathbf{r}_k| = 1.$$

- *ComplEx* [9]:

$$s_{\text{ComplEx}}(h, r, t) = \text{Re}(\langle \mathbf{h}, \mathbf{r}, \bar{\mathbf{t}} \rangle), \quad \mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{C}^d.$$

These scores provide structural priors for ranking candidate actions (r, e') from e_t and are typically learned via link prediction—i.e., training g to rank true triples above corrupted ones to infer missing KG facts.

2.2. General Navigation Model

We model navigation as discrete decisions over the (undirected) *neighborhood* of the current node. Fig. 1 illustrates the question-conditioned navigation setting: a policy walks from the start e_s to an answer e^* , with blue showing one successful path and gray alternatives. At step t the state is $s_t = (e_t, c, t)$, where $e_t \in \mathcal{E}$ is the current entity and c is a *conditioning context*. Let the labeled edges from e_t be

$$\mathcal{N}(e_t) = \{(r, e') : (e_t, r, e') \in \mathcal{T}\}. \quad (1)$$

A scoring function f_θ produces the logits, defining the policy and action space as

$$\pi_\theta(a_t | s_t) = \sigma(f_\theta(e_t, a_t; c)) \quad (2a)$$

$$a_t = (r_t, e_{t+1}), \quad (2b)$$

where σ indicates the softmax operation. Episodes proceed for a fixed hop budget H with $H \geq n$ (where n is the annotated path length); revisits (cycles) are allowed. For a trajectory $\tau = (a_0, \dots, a_{H-1})$ starting at e_s , the log-probability is

$$\log p_\theta(\tau | e_s, c) = \sum_{t=0}^{H-1} \log \pi_\theta(a_t | e_t, c). \quad (3)$$

We denote the *answer set* for an instance by

$$\mathcal{A}(q) \subseteq \mathcal{E}, \quad (4)$$

i.e., the set of entities that are considered correct answers to the query/question q .

Query-conditioned Navigation Task: This is the traditional task of multi-hop in a KG. In the conventional atomic setting, the query q is the relation itself:

$$q := r, \quad c = r, \quad e_s = h,$$

and the answer set is

$$\mathcal{A}(q) = \{t \in \mathcal{E} : (h, r, t) \in \mathcal{T}\}. \quad (5)$$

The policy is conditioned only on r and navigates from h ; to avoid trivial leakage, the target triple (h, r, t) is masked from the immediate choices when constructing the instance.

Question-conditioned Navigation Task: This is our proposed task for the multihop in a KG. To represent natural-language questions, we encode $q = (w_1, \dots, w_m)$ with a pre-trained *language model* (e.g., BERT) to obtain token embeddings $\mathbf{z}_i \in \mathbb{R}^{d_{\text{LM}}}$. We then pool it to a fixed-size vector and *project* it into the KG embedding space $\mathbb{R}^{d_{\text{KG}}}$ (the space of entity/relation embeddings):

$$\mathbf{z}_q = \text{Pool}(\mathbf{z}_1, \dots, \mathbf{z}_m) \in \mathbb{R}^{d_{\text{LM}}}, \\ \tilde{\mathbf{z}}_q = \phi(\mathbf{z}_q) \in \mathbb{R}^{d_{\text{KG}}},$$

where $\phi : \mathbb{R}^{d_{\text{LM}}} \rightarrow \mathbb{R}^{d_{\text{KG}}}$ is a *trainable adapter* that maps the question into the common latent space used by KG embeddings ($\mathbf{e}, \mathbf{r} \in \mathbb{R}^{d_{\text{KG}}}$). We then set the conditioning context and start entity as

$$c = \tilde{\mathbf{z}}_q, \quad e_s \text{ is specified in the instance,}$$

$$\mathcal{A}(q) \subseteq \mathcal{E} \text{ (answer set provided by the dataset).} \quad (6)$$

Here c encodes the full question semantics *in the shared KG-language-model space* used to guide path selection.

2.3. Evaluation: Path Ranking with Multiple Roll-outs

The evaluation is performed by *path ranking*. For each instance (q, e_s) the ranking is obtained as follows:

1. **Rollouts:** Sample $S = 100$ independent trajectories $\{\tau^{(j)}\}_{j=1}^S$ from $\pi_\theta(\cdot | e_s, c)$ with hop budget H .
2. **Compute Score:** Compute the log-score and sort the paths in descending order of $s^{(j)}$.

$$s^{(j)} = \log p_\theta(\tau^{(j)} | e_s, c). \quad (7)$$

3. **Define Success Set and Rank:** Let the terminal node after H hops be $\text{term}(\tau)$ and define

$$\mathcal{J}(q) = \{j \in \{1, \dots, S\} : \text{term}(\tau^{(j)}) \in \mathcal{A}(q)\}. \quad (8)$$

$$\text{rank}(q) = \begin{cases} \min \mathcal{J}(q), & \mathcal{J}(q) \neq \emptyset, \\ S+1, & \mathcal{J}(q) = \emptyset, \end{cases} \quad (9)$$

so that “not found” receives the worst rank $S+1$.

Metrics. We report path-ranking metrics:

$$\text{Hits@}K = \frac{1}{|\mathcal{D}|} \sum_{q \in \mathcal{D}} \mathbf{1}\{\text{rank}(q) \leq K\}, \quad (10)$$

$$\text{MRR}_{\text{path}} = \frac{1}{|\mathcal{D}|} \sum_{q \in \mathcal{D}} \frac{\mathbf{1}\{\text{rank}(q) \leq S\}}{\text{rank}(q)}. \quad (11)$$

We optionally stratify by hop length (e.g., 1/2/3/4 hops) to reflect compositional difficulty. Cycles are permitted; episodes always run for exactly H hops.

3. DATASET

To support *question-conditioned navigation*, we release two QA-KG resources—MQUAKE-ST and KINSHIP—designed for multi-hop navigability and interpretability. Summary statistics appear in Tables 1 and 2. We report: *graph density* (edges as a fraction of all possible edges), *clustering coefficient* (how strongly a node’s neighbors interconnect; higher indicates tighter local cohesion), and the *Krackhardt hierarchy score* on the directed view (proximity to a perfect hierarchy; 1 = fully hierarchical, few cycles). Unlike prior benchmarks such as MetaQA [15], which provide 1–3 hop questions without releasing explicit reasoning paths, and FreebaseQA [16], which primarily offers 1-hop questions, our datasets include full path annotations, specify the start entity, and furnish a materialized (static) KG, enabling path-based navigation and evaluation.

KINSHIP. A toy QA-KG built from the UCI Kinship dataset [17] (distinct from the Australian Kinship dataset [18]). It contains two small family graphs with human-readable person and relation names. Given the limited number of triples, the original validation split is used as the test set. We generate 1–3-hop questions by enumerating simple (acyclic) paths $A \rightarrow \dots \rightarrow B$ in Neo4j, filtering duplicates and invalid relation types, and templating each path into natural language: “Who is the $[r_n]$ of the $\dots$ of the $[r_1]$ of $[A]$?” (answer: B). This dataset serves as a lightweight sandbox; small KG embedding sizes (e.g., $d=12$) are sufficient.

Sample Questions–Answer

1-hop question: “Who is the husband of Penelope?” — *Christopher*
 2-hop Question: “Who is the son of the husband of Penelope?” — *Arthur*
 3-hop Question: “Who is the daughter of the son of the husband of Penelope?” — *Colin*

MQUAKE-ST. Starting from MQuAKE-CF [19]—which provides “True MultiHop” questions with path annotations but no released KG—we materialize an *explicit, static* KG

Table 1. KG statistics (undirected). Counts are triples per split; graph metrics computed on the Full graph (train+valid+test).

Dataset	KINSHIP	MQUAKE-ST
Entities	26	38,278
Relations	13	639
Train triples	104	708,357
Valid triples	32	1,506
Test triples	32	1,506
Avg. node degree	5.23	18.58
Max acyclic path length	2	36
Graph density	4.19e−1	9.71e−4
Clustering coefficient	5.43e−1	1.45e−1
Krackhardt hierarchy	1.00	9.71e−1

Table 2. Question counts by hop length (from start entity).

Dataset	KINSHIP	MQUAKE-ST
1-Hop	112	-
2-Hop	248	4,879
3-Hop	420	2,535
4-Hop	-	1,804
Total	780	9,218

for navigation, MQUAKE-ST. It is a general-knowledge resource aligned to Wikidata QIDs with human-readable labels, enabling interoperability and reproducibility. “Static” means we discard MQUAKE’s edited triples (and their altered answers) and retain only the original triples; questions are used unmodified. To avoid the sparsity of a path-only graph, we (i) seed nodes/relations/triples from the QA paths, (ii) augment it via Wikidata, and (iii) filter uncommon/noisy edges, then freeze the result as a static KG.

Sample Questions–Answer

1-hop question: “Who founded Let’s Move!?” — *Michelle Obama*

2-hop question: “Who is the partner of the founder of Let’s Move!?” — *Barack Obama*

3-hop question: “From which country does the partner of the founder of Let’s Move! originate?” — *United States*

4. EXPERIMENTS

Protocol. We use the path-ranking evaluation of Sec.2.3 with 100 roll-outs per question. Episodes run for exactly H hops, where H equals the annotated hop count of the question. The KG is treated as undirected, cycles are allowed, and the per-step action set is truncated to at most 200 candidates. KG

Table 3. Navigation performance on KINSHIP and MQUAKE-ST. RW rows report *arrival probability*; learned agents report *Hits@1* $\uparrow$. Columns are QA hop buckets; “Full” uses train+valid+test edges, “Train” uses train-only edges. All results use 100 roll-outs with a fixed hop budget H .

KINSHIP					
Model/ QA Hop Size	Graph	1-Hop $\uparrow$	2-Hops $\uparrow$	3-Hops $\uparrow$	n-Hops $\uparrow$
RW-End	Full	1.72e−1	8.19e−2	7.41e−2	8.13e−2
	Train	8.11e−2	7.46e−2	7.09e−2	7.43e−2
RW-Gold	Full	8.59e−2	7.11e−3	6.04e−4	3.55e−3
	Train	8.13e−1	8.39e−1	7.69e−1	7.58e−1
MINERVA	Full	3.44e−1	5.97e−1	6.73e−1	4.44e−1
	Train	8.13e−1	8.39e−1	7.69e−1	7.58e−1
MQUAKE-ST					
Model/ QA Hop Size	Graph	2-Hop $\uparrow$	3-Hops $\uparrow$	4-Hops $\uparrow$	n-Hops $\uparrow$
RW-End	Full	8.53e−3	3.61e−3	1.55e−3	3.39e−3
	Train	8.53e−3	3.61e−3	1.56e−3	3.40e−3
RW-Gold	Full	1.55e−3	6.69e−6	5.00e−8	3.26e−5
	Train	7.03e−1	3.94e−1	9.01e−1	8.12e−1
MINERVA	Full	8.61e−1	8.39e−1	9.50e−1	8.25e−1
	Train	4.33e−1	4.82e−1	3.55e−1	2.96e−1
MultiHopKG	Full	3.72e−1	3.51e−1	3.26e−1	3.47e−1
	Train	8.38e−1	8.68e−1	6.64e−1	8.35e−1
SQUIRE	Full	4.99e−1	8.15e−1	6.74e−1	6.75e−1
	Train	4.99e−1	8.15e−1	6.74e−1	6.75e−1

embedding dimensions are $d_{\text{KG}}=12$ for KINSHIP (toy scale) and $d_{\text{KG}}=100$ for MQUAKE.

Hyperparameters. All hyperparameters are shared across hop buckets; only the hop budget H varies. For mixed-hop buckets (e.g., 2–4), we fix H to the bucket maximum (e.g., $H=4$). No hyperparameter tuning is done.

Baselines. Besides adapted MINERVA/MultiHopKG/SQUIRE, we report two random-walk (RW) baselines on the undirected graph: (i) **RW-End**: an H -step unbiased walk from e_s succeeds if the terminal node lies in $\mathcal{A}(q)$; (ii) **RW-Gold**: the probability of exactly following the annotated path $\mathcal{P}=(e_0, \dots, e_H)$, computed as $\prod_{t=0}^{H-1} 1/\deg(e_t)$.

Reporting. We report Hits@1 under two graph regimes: *Complete KG* (train+valid+test triples as edges) and *Incomplete KG* (train-only edges). Results are summarized in Table 3.

Results. On KINSHIP, an unbiased walk that “lands anywhere correct” (RW-End) yields a non-trivial floor (17% for 1-hop; $\sim 8\%$ for 2–3 hops), while exactly following the gold path (RW-Gold) collapses with hop length ($8.6\% \rightarrow 0.71\% \rightarrow 0.06\%$). Learned agents comfortably exceed these floors (e.g., MINERVA 0.81/0.84/0.77 on the Full graph) but degrade under Train-only edges, indicating sensitivity to missing links. Additionally, shorter paths are most sensitive to edge availability: on KINSHIP, MINERVA drops sharply at 1-hop ($0.81 \rightarrow 0.34$), yet the Full–Train gap narrows by 3 hops ($0.77 \rightarrow 0.67$). Results for Squire and MultiHopKG are omitted for brevity, as they highlight similar performance and provide no additional insight.

On MQUAKE-ST, random walks are essentially ineffective (RW-End $< 1\%$ and RW-Gold ≈ 0), further highlighting differences in performance between methods. MINERVA shows strong performance and, unlike on KINSHIP, is more robust to missing edges: using the Train-only graph it consistently tops its Full-graph scores across hop buckets and slightly in aggregate. SQUIRE’s performance is strongest on the augmented Full graph but drops when edges are removed, indicating sensitivity to sparsity. MultiHopKG trails overall; nonetheless, on the Full graph it edges out MINERVA in the 3-hop bucket, and it maintains a steadier aggregate on Train than on Full, hinting at some tolerance to pruning.

Takeaway. Random success is non-trivial in small toy graphs but effectively vanishes in realistic KGs; consequently, effective navigation requires informative policies rather than chance. Missing edges primarily degrade short-hop questions, whereas KG augmentation—by increasing branching—often reduces path fidelity for step-wise walkers but can improve path-planning models that exploit the additional connectivity.

Code, models, and results. We release adapted navigation code (MINERVA, MultiHopKG, SQUIRE), evaluation scripts, and extended metrics (Hits@1/3/5/10, MRR) at HalcyonSolutions; see MINERVA, MultiHopKG, and SQUIRE.

Data. The datasets are hosted at KINSHIP and MQUAKE-ST.

5. REFERENCES

- [1] Yilin Wen, Zifeng Wang, and Jimeng Sun, “Mindmap: Knowledge graph prompting sparks graph of thoughts in large language models,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 10370–10388.
- [2] Wooyoung Kim, Haemin Jung, and Wooju Kim, “Knowledge graph as pre-training corpus for structural reasoning via multi-hop linearization,” *IEEE Access*, 2024.
- [3] Junnan Dong, Qinggang Zhang, Xiao Huang, Keyu Duan, Qiaoyu Tan, and Zhimeng Jiang, “Hierarchy-aware multi-hop question answering over knowledge graphs,” in *Proceedings of the ACM web conference 2023*, 2023, pp. 2519–2527.
- [4] Zijian Qiao, Nan Li, Chenxi Huang, Gangliang Wang, Shenglin Liang, Hui Lin, and Qinglang Guo, “Graphllm: A general framework for multi-hop question answering over knowledge graphs using large language models,” in *CCF International Conference on Natural Language Processing and Chinese Computing*. Springer, 2024, pp. 136–148.
- [5] Abir Chakraborty, “Multi-hop question answering over knowledge graphs using large language models,” *arXiv preprint arXiv:2404.19234*, 2024.
- [6] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko, “Translating embeddings for modeling multi-relational data,” *Advances in neural information processing systems*, vol. 26, 2013.
- [7] Zhiqing Sun, Zhi-Hong Deng, Jian-Yun Nie, and Jian Tang, “Rotate: Knowledge graph embedding by relational rotation in complex space,” *arXiv preprint arXiv:1902.10197*, 2019.
- [8] Bishan Yang, Scott Wen-tau Yih, Xiaodong He, Jianfeng Gao, and Li Deng, “Embedding entities and relations for learning and inference in knowledge bases,” in *Proceedings of the International Conference on Learning Representations (ICLR) 2015*, 2015.
- [9] Théo Trouillon, Johannes Welbl, Sebastian Riedel, Éric Gaussier, and Guillaume Bouchard, “Complex embeddings for simple link prediction,” in *International conference on machine learning*. PMLR, 2016, pp. 2071–2080.
- [10] Tim Dettmers, Pasquale Minervini, Pontus Stenetorp, and Sebastian Riedel, “Convolutional 2d knowledge graph embeddings,” in *Proceedings of the AAAI conference on artificial intelligence*, 2018, vol. 32.
- [11] Wenhan Xiong, Thien Hoang, and William Yang Wang, “Deep-path: A reinforcement learning method for knowledge graph reasoning,” in *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 2017, pp. 564–573.
- [12] Rajarshi Das, Shehzaad Dhuliawala, Manzil Zaheer, Luke Vilnis, Ishan Durugkar, Akshay Krishnamurthy, Alex Smola, and Andrew McCallum, “Go for a walk and arrive at the answer: Reasoning over paths in knowledge bases using reinforcement learning,” *arXiv preprint arXiv:1711.05851*, 2017.
- [13] Xi Victoria Lin, Richard Socher, and Caiming Xiong, “Multi-hop knowledge graph reasoning with reward shaping,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 3243–3253.
- [14] Yushi Bai, Xin Lv, Juanzi Li, Lei Hou, Yincen Qu, Zelin Dai, and Feiyu Xiong, “Squire: a sequence-to-sequence framework for multi-hop knowledge graph reasoning,” *arXiv preprint arXiv:2201.06206*, 2022.
- [15] Yuyu Zhang, Hanjun Dai, Zornitsa Kozareva, Alexander J Smola, and Le Song, “Variational reasoning for question answering with knowledge graph,” in *AAAI*, 2018.
- [16] Kelvin Jiang, Dekun Wu, and Hui Jiang, “Freebaseqa: A new factoid qa data set matching trivia-style question-answer pairs with freebase,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 318–323.
- [17] Geoff Hinton, “Kinship,” UCI Machine Learning Repository, 1986, DOI: <https://doi.org/10.24432/C5WS4D>.
- [18] Woodrow W Denham, Chad K McDaniel, and John R Atkins, “Aranda and alyawara kinship: A quantitative argument for a double helix model,” *American Ethnologist*, vol. 6, no. 1, pp. 1–24, 1979.
- [19] Zexuan Zhong, Zhengxuan Wu, Christopher D Manning, Christopher Potts, and Danqi Chen, “Mquake: Assessing knowledge editing in language models via multi-hop questions,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 15686–15702.